

Data Acquisition

Raw MBP-10 parquet
~7M tick events per symbol
6 symbols · 4 trading days

Offline Feature Research

Chronological split
train_size rows

Feature pipeline
fit on train · transform both splits
Welford online normalisation

IC / ICIR scoring
rolling Spearman rank correlation
vs Sharpe-proxy target
horizons [10, 50, 200]

Aggregate across symbols
mean ICIR ranking

Greedy conditional IC selection
linear residualisation
for redundancy pruning

selected_features.yaml
reduced feature config

Data Preparation

Chronological split per symbol
train · val · test

Feature pipeline
fit on train only
Welford online normalisation

Parquet cache
data/prepared/
train · val · test per symbol

Memmap files
data/memmap/
{i}_train_data.npy float32
{i}_train_index.npy int64
{i}_columns.json

memmap paths
per symbol

RL Training

StreamingTradingEnvXY
reset(): shuffle symbol deck
pick random 50k-row window
from that symbol's memmap

SyncDataCollector
collect frames_per_batch=200
transitions per update

Replay Buffer
LazyTensorStorage
buffer_size=100k transitions

TD3Loss
twin critics · clipped double Q
delayed actor update (every 2 steps)
target policy smoothing $\sigma=0.2$

Soft target network update
 $\tau = 0.001$

Checkpoint
actor + twin critic weights
optimizer state
every 100k steps

Evaluation

Deterministic rollout
no exploration noise
full val split · full test split

Per-split metrics
Sharpe ratio · max drawdown
total return · win rate
position change ratio

Statistical significance tests
t-test · Sharpe bootstrap
Sortino bootstrap · Mann-Whitney
vs buy-and-hold · TWAP · VWAP · random

Explainability
permutation feature importance
n_steps=500

MLflow
parameters · metrics
artifacts · plots

feeds reduced
feature config

val + test
DataFrames

next batch

trained
actor weights